

AWS CRUD - PDF técnico

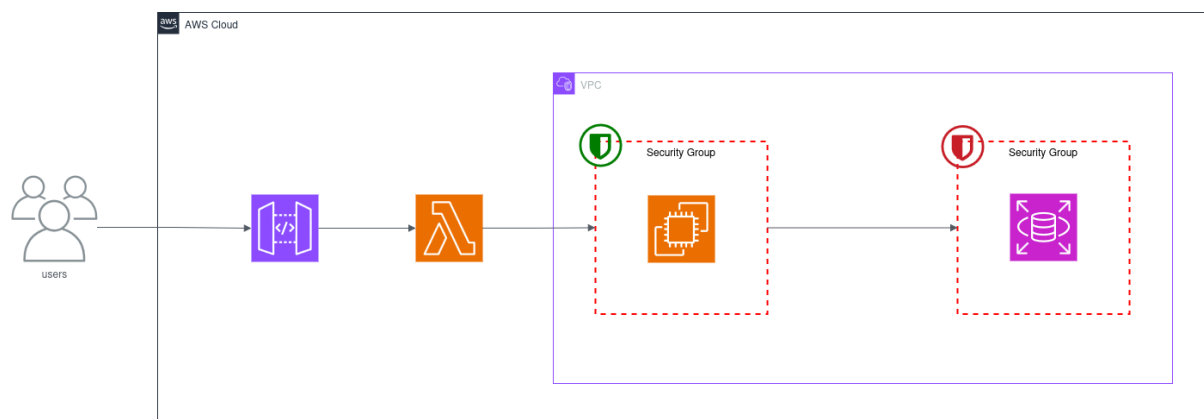
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Neste documento vamos declarar a arquitetura do nosso sistema, assim como os serviços utilizados

1- Diagrama do sistema

O nosso sistema utiliza o API Gateway para direcionar todas as requisições em determinado domínio para a nossa função lambda, que por sua vez requisita um backend instanciado em uma ec2, o qual acessa dados para em banco de mysql no rds, e com o retorno do backend, a função lambda retorna algumas estatísticas.

O banco de dados pode ser acessado apenas pelo backend, porém o backend assim como a função lambda podem ser acessados por qualquer ip.



2- Capturas de tela dos serviços

2.1 - API Gateway

The screenshot displays the AWS API Gateway console. The top section, titled 'APIs (1/1)', contains a search bar and a table with one entry: 'nba-api-gateway' (ID: d9x692bnpf, Protocol: HTTP, API endpoint type: Regional, Created: 2025-05-23). Below this is the 'Integrations' section, which has two tabs: 'Attach integrations to routes' and 'Manage integrations'. The 'Attach integrations to routes' tab shows a tree view of routes for 'nba-api-gateway', with the selected route being 'GET /report' using the 'AWS Lambda' provider. The 'Manage integrations' tab shows the 'Integration details for route' for 'GET /report (tmtins)'. The details include the 'Lambda function' 'nbaFunction (us-east-1)', the 'Integration ID' 't68qx17', and sections for 'Payload format version' (2.0), 'Invoke permissions', 'Example policy statement', 'Timeout' (30000), 'Request parameter mapping', and 'Response parameter mappings'.

APIs (1/1)

Find APIs

Name	Description	ID	Protocol	API endpoint type	Created
nba-api-gateway		d9x692bnpf	HTTP	Regional	2025-05-23

Integrations

Attach integrations to routes | Manage integrations

Routes for nba-api-gateway

Search

- ▼ /nbaFunction
 - ANY
 - ▼ /report
 - GET **AWS Lambda**

Integration details for route

GET /report (tmtins)

Lambda function
nbaFunction (us-east-1)

Integration ID
t68qx17

Description
-

Payload format version
The parsing algorithm for the payload sent to and returned from your Lambda function. [Learn more](#)
2.0 (Interpreted response format)

Invoke permissions
The resource policy of the Lambda function determines if API Gateway can invoke it. You can run the AWS CLI command snippet below to give API Gateway permission to invoke your AWS Lambda function.

► **Example policy statement**

Timeout
The number of milliseconds that API Gateway should wait for a response from the integration before timing out.
30000

Request parameter mapping
Not configured

Response parameter mappings
Not configured

2.2 - Lambda

Functions (6)

Last fetched 10 seconds ago

Actions

Create function

Filter by attributes or search by keyword

☐	Function name	Description	Package type	Runtime	Last modified
☐	MainMonitoringFunction	-	Zip	Python 3.9	3 days ago
☐	ModLabRole	updates LabRole to allow it to assume itself	Zip	Python 3.9	3 days ago
☐	RedshiftOverwatch	Deletes Redshift Cluster if the count is more than 2.	Zip	Python 3.9	3 days ago
☐	RedshiftEventSubscription	Create Redshift event subscription to SNS Topic.	Zip	Python 3.9	3 days ago
☐	RoleCreationFunction	Create SLR if absent	Zip	Python 3.9	3 days ago
☐	nbaFunction	-	Zip	Python 3.13	34 minutes ago

nbaFunction

Throttle

Copy ARN

Actions

Function overview

Diagram

Template

nbaFunction

Layers

(0)

API Gateway

+ Add trigger

+ Add destination

Export to Infrastructure Composer

Download

Description

-

Last modified

34 minutes ago

Function ARN

arn:aws:lambda:us-east-1:279381601805:function:nbaFunction

Function URL

-

Code

Test

Monitor

Configuration

Aliases

Versions

Code source

Upload from

EXPLORER

▼ NBFUNCTION

urlib3

urlib3-2.4.0.dist-info

lambda_function.py

requirements.txt

▼ DEPLOY

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

▼ TEST EVENTS [SELECTED: NBFUNCTION]

+ Create new test event

Private saved events

nbaFunction

▼ ENVIRONMENT VARIABLES

Amazon Q

lambda_function.py

```

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def lambda_handler(event, context):
    url = "http://ec2-54-158-72-226.compute-1.amazonaws.com"
    response = requests.get(url)
    data = response.json()

    teams_total = len(data)
    teams_by_city = {}
    oldest_team = None
    oldest_team_date = None
    for team in data:
        city = team['cidade']
        if city not in teams_by_city:
            teams_by_city[city] = 0
        teams_by_city[city] += 1
        if oldest_team_date is None or datetime.strptime(
            oldest_team_date, '%Y-%m-%d') > datetime.strptime(team['dataFundacao'], '%Y-%m-%d'):
            oldest_team_date = team['dataFundacao']
            oldest_team = team['nome']

    oldest_team_city = team['cidade']
    oldest_team_date = team['dataFundacao']
    city_with_most_teams = max(teams_by_city, key=teams_by_city.get)
    city_with_most_teams_count = teams_by_city[city_with_most_teams]

    response = {
        "total_teams": teams_total,
        "cidade_com_mais_times": {
            "cidade": city_with_most_teams,
            "quantidade": city_with_most_teams_count
        },
        "time_mais_antigo": {
            "nome": oldest_team,
            "dataFundacao": oldest_team_date
        }
    }

    return response

```

requirements.txt

requirements.txt

requirements.txt

Ln 9, Col 1 Spaces: 4 UTF-8 LF Python Lambda Layout: us

✓ Executing function: succeeded ([logs](#))

▼ Details

```
{
  "statusCode": 200,
  "body": "{\"total_times\": 5, \"cidade_com_mais_times\": {\"cidade\": \"S\u00e3o Paulo\", \"quantidade\": 1}, \"time_mais_antigo\": {\"nome\": \"Le\u00f5es do Cerrado\", \"dataFundacao\": \"Mon, 10 Jul 1995 00:00:00 GMT\", \"cidade\": \"Bras\u00edlia\"}}"
```

Summary

Code SHA-256 MBHYHQUL1/hD/bFE8QgFipCjaFSTihzCTEw+4khfRA=	Execution time 1 second ago
Function version \$LATEST	Request ID 7e2dc400-7e05-4b8f-bb84-b56b9b6431f6
Duration 124.09 ms	Billed duration 125 ms
Resources configured 128 MB	Max memory used 63 MB
Init duration 370.53 ms	

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

```
START RequestId: 7e2dc400-7e05-4b8f-bb84-b56b9b6431f6 Version: $LATEST
END RequestId: 7e2dc400-7e05-4b8f-bb84-b56b9b6431f6
REPORT RequestId: 7e2dc400-7e05-4b8f-bb84-b56b9b6431f6  Duration: 124.09 ms    Billed Duration: 125 ms Memory Size: 128 MB    Max Memory Used: 63 MB Init Duration: 370.53 ms
```

✓ Executing function: succeeded ([logs](#))

► Details

Test event Info Delete CloudWatch Logs Live Tail Save Test

To invoke your function without saving an event, modify the event, then choose Test. Lambda uses the modified event to invoke your function, but does not overwrite the original event until you choose Save.

Test event action

☐ Create new event ☒ Edit saved event

Event name

nbaFunction

Event JSON Format JSON

```
1 {
2   }
```

1/2 JSON

2.3 - EC2

Instances (1) Info

Find instance by attribute or tag (case-sensitive)

All states

Connect

Instance state

Actions

Launch instances

<1>

Settings

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
nba-app	i-08364763c00d69df5	Running	t2.micro	2/2 checks passed	View alarms	us-east-1b	ec2-54-158-72-226.co...	54.158.72.226	-	-

Instance summary for i-08364763c00d69df5 (nba-app) Info

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID

i-08364763c00d69df5

IPv6 address

-

Hostname type

IP name: ip-172-31-88-245.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

54.158.72.226 [Public IP]

IAM Role

-

IMDSv2

Required

Operator

-

Public IPv4 address

54.158.72.226 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-88-245.ec2.internal

Instance type

t2.micro

VPC ID

vpc-0482aa6373f1511f

Subnet ID

subnet-0acbcd1cae57186b3

Instance ARN

arn:aws:ec2:us-east-1:279381601805:instance/i-08364763c00d69df5

Private IPv4 addresses

172.31.88.245

Public DNS

ec2-54-158-72-226.compute-1.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance details Info

AMI ID

ami-0953476d60561c955

AMI name

al2023-ami-2023.7.20250512.0-kernel-6.1-x86_64

Stop protection

Disabled

Instance auto-recovery

Default

AMI Launch index

0

Credit specification

standard

Usage operation

RunInstances

Enclaves Support

-

Allow tags in instance metadata

Disabled

▼ Host and placement group Info

Host ID

-

Host resource group name

-

Virtualization type

hvm

Number of vCPUs

1

Monitoring

disabled

Allowed image

-

Launch time

Fri May 23 2025 15:50:34 GMT-0300 (Brasilia Standard Time) (about 1 hour)

Lifecycle

normal

Key pair assigned at launch

nbaApp

Kernel ID

-

RAM disk ID

-

Boot mode

uefi-preferred

Use RBN as guest OS hostname

Disabled

Platform details

Linux/UNIX

Termination protection

Disabled

AMI location

amazon/al2023-ami-2023.7.20250512.0-kernel-6.1-x86_64

Stop-hibernate behavior

Disabled

State transition reason

-

State transition message

-

Owner

279381601805

Current instance boot mode

legacy-bios

Answer RBN DNS hostname IPv4

Enabled

Affinity

-

Tenancy

default

Reservation

r-0083a68e630b6003f

Placement group

-

Placement group ID

-

Partition number

-

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM Role

-

Security groups

sg-0d9d377bb52be9b94 (nba-app)

Owner ID

279381601805

Launch time

Fri May 23 2025 15:50:34 GMT-0300 (Brasilia Standard Time)

▼ Inbound rules

Filter rules

<1>

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-092a117f155936490	80	TCP	0.0.0.0/0	nba-app	-
-	sgr-0aca28b8034c92f57	30000	TCP	0.0.0.0/0	nba-app	-
-	sgr-04626ada3dbf13cb2	22	TCP	0.0.0.0/0	nba-app	-
-	sgr-067d6e05ecd113238	443	TCP	0.0.0.0/0	nba-app	-

▼ Outbound rules

Filter rules

<1>

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
-	sgr-092daa108170e3f6c	All	All	0.0.0.0/0	nba-app	-


```

~ $ curl http://ec2-54-158-72-226.compute-1.amazonaws.com:30000/times;
[
  {
    "cidade": "S\u00e3o Paulo",
    "dataFundacao": "Sun, 15 Apr 2001 00:00:00 GMT",
    "id": "1",
    "nome": "Lobos de S\u00e3o Paulo",
    "tecnico": "Carlos Mendes"
  },
  {
    "cidade": "Porto Alegre",
    "dataFundacao": "Mon, 21 Sep 1998 00:00:00 GMT",
    "id": "2",
    "nome": "\u00c1guias do Sul",
    "tecnico": "Fernanda Lima"
  },
  {
    "cidade": "Rio de Janeiro",
    "dataFundacao": "Sat, 08 Jan 2005 00:00:00 GMT",
    "id": "3",
    "nome": "Tigres da Vila",
    "tecnico": "Jo\u00e3o Silva"
  },
  {
    "cidade": "Manaus",
    "dataFundacao": "Tue, 30 Nov 2010 00:00:00 GMT",
    "id": "4",
    "nome": "Falc\u00f5es do Norte",
    "tecnico": "Patr\u00edcia Rocha"
  },
  {
    "cidade": "Bras\u00edlia",
    "dataFundacao": "Mon, 10 Jul 1995 00:00:00 GMT",
    "id": "5",
    "nome": "Le\u00f5es do Cerrado",
    "tecnico": "Marcos Tavares"
  },
  {
    "cidade": "Chicago",
    "dataFundacao": "Sun, 16 Jan 1966 00:00:00 GMT",
    "id": "6",
    "nome": "Chicago Bulls",
    "tecnico": "Billy Donovan"
  }
]
~ $ █

```

```

~ $ curl -X PUT http://ec2-54-158-72-226.compute-1.amazonaws.com:30000/times/6 \
> -H "Content-Type: application/json" \
> -d '{
>   "nome": "Chicago Bulls Atualizado",
>   "cidade": "Chicago",
>   "dataFundacao": "1966-01-16",
>   "tecnico": "Novo T\u00e9cnico"
> }';
{
  "mensagem": "Time atualizado"
}
~ $ █

```

```
~ $ curl -X DELETE http://ec2-54-158-72-226.compute-1.amazonaws.com:30000/times/6;
{
  "mensagem": "Time deletado"
}
~ $
```

```
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
14986ea82833   nba-app   "python app.py"         2 minutes ago Up 2 minutes   0.0.0.0:30000->30000/tcp, :::30000->30000/tcp   practical_hellman
[ec2-user@ip-172-31-88-245 nba-crud]$ docker logs 14986ea82833
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:30000
* Running on http://172.17.0.2:30000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 133-542-428
3.87.239.25 - - [23/May/2025 20:31:39] "POST /times HTTP/1.1" 201 -
3.87.239.25 - - [23/May/2025 20:32:29] "POST /times HTTP/1.1" 415 -
3.87.239.25 - - [23/May/2025 20:32:44] "GET /times HTTP/1.1" 200 -
3.87.239.25 - - [23/May/2025 20:33:19] "PUT /times/6 HTTP/1.1" 200 -
3.87.239.25 - - [23/May/2025 20:33:47] "DELETE /times/6 HTTP/1.1" 200 -
[ec2-user@ip-172-31-88-245 nba-crud]$
```

2.4 - RDS

Databases (1)
Group resources
Modify
Actions
Create database

Filter by databases

DB Identifier	Status	Role	Engine	Region ...	Size	Recommendations
nba-database	Available	Instance	MySQL Co...	us-east-1a	db.t4g.micro	2 Informational

nba-database
Modify
Actions

Summary

DB Identifier
nba-database

Status
Available

Role
Instance

Engine
MySQL Community

Recommendations
2 Informational

CPU
3.17%

Class
db.t4g.micro

Current activity
0 Connections

Region & AZ
us-east-1a

Connectivity & security
Monitoring
Logs & events
Configuration
Zero-ETL integrations
Maintenance & backups
Data migrations - new
Tags
Recommendations

Connectivity & security

Endpoint & port

Endpoint
nba-database.c0jdr9hfgdy.us-east-1.rds.amazonaws.com

Port
3306

Networking

Availability Zone
us-east-1a

VPC
vpc-0482aa6373f1511f

Subnet group
default-vpc-0482aa6373f1511f

Subnets
subnet-0407b61abe8cb01e4
subnet-045cc1f4fed20b1d4
subnet-0093d4a493bc2feb5
subnet-0acbcf1cae57186b5
subnet-0a9e0be9dbb1526c5
subnet-07e00954ac1303b01

Network type
IPv4

Security

VPC security groups
my-db-sec-group (sg-018805844544a88c9)
Active

Publicly accessible
No

Certificate authority
rds-ca-rsa2048-g1

Certificate authority date
May 25, 2061, 20:34 (UTC-03:00)

DB instance certificate expiration date
May 20, 2026, 23:21 (UTC-03:00)

3- Atividades de cada membro

Membro	Atividade
Luiz	Integração da api com o banco de dados
Joao Victor	Criação do API Gateway
Pedro	Desenvolvimento da api
Matheus	Criação da instância EC2
Yuri	Criação da instância RDS